

Post-Pandemic Gender Labor Force Trends

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The study of women's participation in the American workforce is not a groundbreaking research topic in the Economic discipline. The transition of women from homemaking to job taking in the World War 2 era, then the sharp fall backwards during the COVID-19 pandemic is an interesting ongoing transition to analyze. However, a new trend has emerged: the labor force participation rate of men is decreasing. According to the Bureau of Labor Statistics, "During the 1996–2016 period, the nonparticipation rate increased the most for younger men of prime working age, those age 25 to 34" (BLS). Seeing as this counters traditional gender norms of the 'bread winning' man, it sparks interesting questions about its cause and what it says about our greater society. Therefore, in this project, we will ask: **Is the labor force participation rate of men really decreasing after the COVID-19 pandemic? If so, what is causing this trend? Will these trends persist?**

Our study took multiple approaches, investigating if either the age distribution of men and women or the industries they choose could be the answer to our questions. Our study began with a time series analysis, in which we determined if there truly is a difference in labor force participation by gender. A double-difference regression allowed us to quantify that labor force participation rate after COVID recovered by 0.99% more for women than for men, considering a 5-year window before and after COVID. We then ran an interrupted time series analysis by industry and a difference in difference by age, to evaluate the significance of either of those variables. Finally, we ran a regression controlling for age group to see if age could explain the trend the best. This leads us to conclude that **the post pandemic return of women aged 25-44 and industries they tend to be a part of, are the main explanatory variables for this gap.**

The Data

We began our research by cleaning the data. The file, a historical record of labor force time series data from the Bureau of Labor Statistics, needed to be split into three dataframes: one containing the labor force participation rates by gender, one containing labor force participation rates by age and gender, and one containing total employment in each industry by gender. The data was converted to datetime objects, and ensured that men, women, and each industry had a data point for each month along the series. Other cleaning, filtering, and indexing was also conducted.

The Analysis

1. Initial Autoregression

Then an initial time series analysis was performed. An autoregression was run on both the men and women's labor force participation rates over time. One particular challenge we encountered was the appropriate number of lags to consider. The original time series was proven to be non-stationary, meaning certain properties like mean and variance changed over time. This was confirmed with an Augmented Dickey-Fuller (ADF) test, which gave a high p-value (around 0.78), so we failed to reject the null hypothesis of non-stationarity, which causes misleading results in time series models. Therefore, we took the first difference of the series instead, looking at the month-to-month changes instead of the raw data trend over time. This transforms the data into a stationary series suitable for modeling.

We then used an autoregression of both men's and women's data (with the determined 1 lags) to forecast the trend into the future. This allowed us to answer the question of whether there is a trend of declining male participation and if it will continue, or if it is just an economic phase. The predictions, once plotted, did not have a stark trend.

The comparison of month-to-month changes overall suggests that neither gender has a pattern of being more above or below zero [Figure 1]. The forecasted lines, seen in the red line above the black, suggest that females have a slightly more positive trend.

What is significant is the sharp dip around 2020, demonstrating the dramatic decrease in participation that followed the COVID-19 pandemic. When the visualization is zoomed to the timeframe surrounding the pandemic, 2000-2032 [Figure 2] , it becomes clearer that the initial shutdown of the pandemic was associated with significant declines in labor force participation for both men and women in similar proportions. What happens after the pandemic is difficult to discern, as this timeseries trend of monthly changes does not make it easy to conclude if men or women's participation rate comes back stronger post-pandemic.

2. Double Difference Regression

Therefore, using a dataset containing each month's labor force participation rate for men and women from five years before COVID to five years after, we ran a regression of labor force participation on the variable "female," which is 1 when the labor force participation rate is for women, and the variable "post," which is 1 when the labor force participation rate is from a date after COVID. The coefficient on the interaction term of female and post is a statistically significant 0.99. This coefficient is the double difference between men and women and pre and post COVID. It signifies that the labor force participation rate after COVID recovered by 0.99% more for women than for men. [Figure 3]

After establishing a gender difference in post-COVID labor force trends, we asked what is causing this difference? The next logical step was to investigate the difference in labor force participation of genders controlling for different variables. One of those variables was difference by industry.

3. Analysis by Industry

To do so, we ran an interrupted time series analysis with regression of employment totals on time, relative post-COVID year, and categorized industry variables based on employment data from five years before COVID to five years after. We then filtered for the industries with a statistically significant p-value (less than .05) for the time since COVID variables which revealed industries such as manufacturing, construction, and education and health services [Table 1]. COVID had a significant negative long term effect on the trend of total employment of these 14 industries, which means that they were not able to bounce back like the 16 other industries that had a statistically significant positive time since COVID p-value in the data set (when controlling for different occupations as a fixed effect). This could potentially be logical, in that manufacturing and construction industries are traditionally dominated by men.

We also looked at the 16 industries that grew significantly after COVID. These included accommodation and food service, arts and entertainment, educational services, and health care [Table 2]. This also could potentially make logical sense, in that they are traditionally dominated by women.

Although these 14 industries explain some of the male downward trend, it likely does not explain the whole story. Therefore, it made sense to consider a second variable.

5. Analysis by Age

Next, we regressed labor force participation on gender controlling for age. A facet plot of labor force participation by age and gender demonstrates the results [Figures 4 & 5]. It revealed that **women ages 25-44 contribute to the gap in gender the most. They experienced a large and unique rise in labor force participation.** Compared with men of the same age, women 25-34 and 35-44 saw roughly a +2.1 to +2.3 percentage-point additional increase in labor force participation after COVID. This increase is statistically significant, and larger than the results for younger or older women. Younger women, ages 16-24, did see a recovery, but not as large as the middle age group. Older women, ages 45 and older, also saw a very small increase compared to men, but was not statistically significant.

6. Conclusion and Further Research

In conclusion, by examining the overall timeseries of the data, and comparing male and female pre and post pandemic labor force participation rates, it was obvious that there was a difference in gender labor trends, but it was not clear what. That led us to conduct a regression of change in employment by industry, which explained some of the answers, but still left some questions. We then ran a difference-in-difference study by

age group, to investigate whether age explained this trend. **Our results of the individual industry and age analysis agree and are logical: middle-aged women, ages 25-44, demonstrate the highest growth in labor force participation relative to men, and tend to participate in the industries that saw the most growth, such as educational services, and health care.** There is statistically significant data to suggest that both of those things are true, and they happen congruently.

It is logical that these particular industries increased after the pandemic. Teachers, which fall under educational services, faced increased difficulties in their jobs during the pandemic, due to virtual teaching of students and the increased stress associated with doing so. According to the Brookings Institute, “a RAND survey, fielded in early January 2021, found that nearly one-quarter of teachers indicated a desire to leave their jobs at the end of the school year, compared with an average national turnover rate of 16% pre-pandemic according to NCES data”. Therefore, after the pandemic, teachers may have felt willing and able to return to the field. Moreover, according to PEW Research “Most K-12 public school teachers are women. About three-quarters (77%) of teachers” are female. The culmination of these facts and our data add to a sound conclusion, that post college age women leaving their teaching positions during the pandemic and returning afterwards may explain this trend in the gap.

Similarly, many healthcare professionals may have left their jobs during the pandemic due to health risks of themselves or their families, and returned once it was safe again. Likewise, nurses in these positions tend to be women. According to National University, “Women continue to account for a large majority of nurses. Currently, men account for 11% of the RN workforce,” which implies that women account for the other 89%. This also matches our data findings.

There were some limitations to our research, in that we did not have a data set that included labor force participation by industry, due to the fact that by default, not participating in the labor force means you are not in an industry. Further research will also have to look into the age decomposition of industries.

7. Other Research and Literature

Other literature and scholarship has been conducted on this topic. According to Villarreal and Yu’s short term fixed effects model, published in the National Library of Medicine in 2022, revealed that when compared to men’s labor force participation, **“women's full-time employment did not decline significantly relative to that of men during the months following the beginning of the epidemic”**. They also controlled for variables that we did not, such as the presence of children and seasonality of employment. If we had the data to control for these issues, we may have seen similar results.

Appendix

Figure 1:

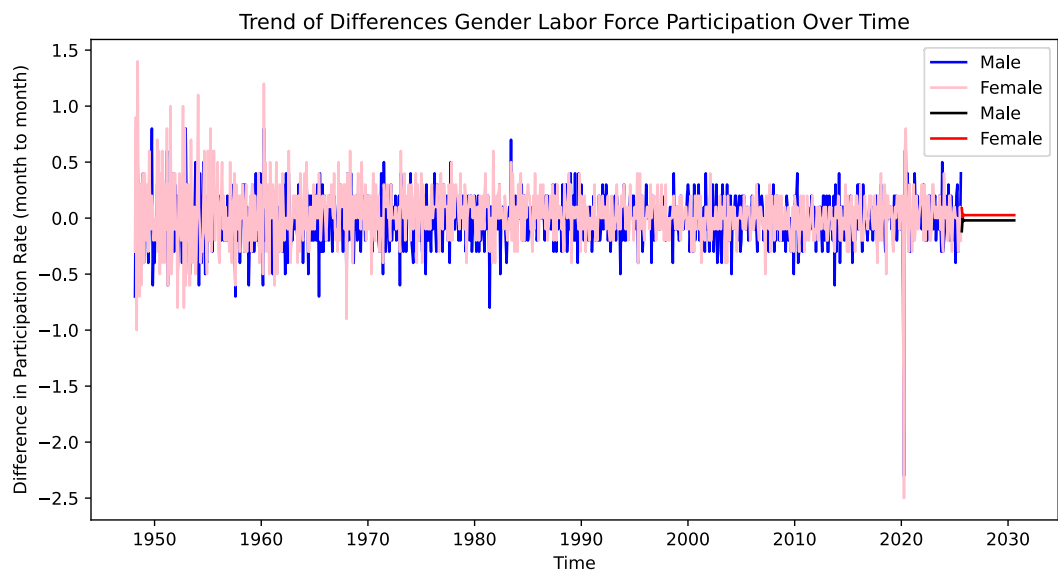


Figure 2:

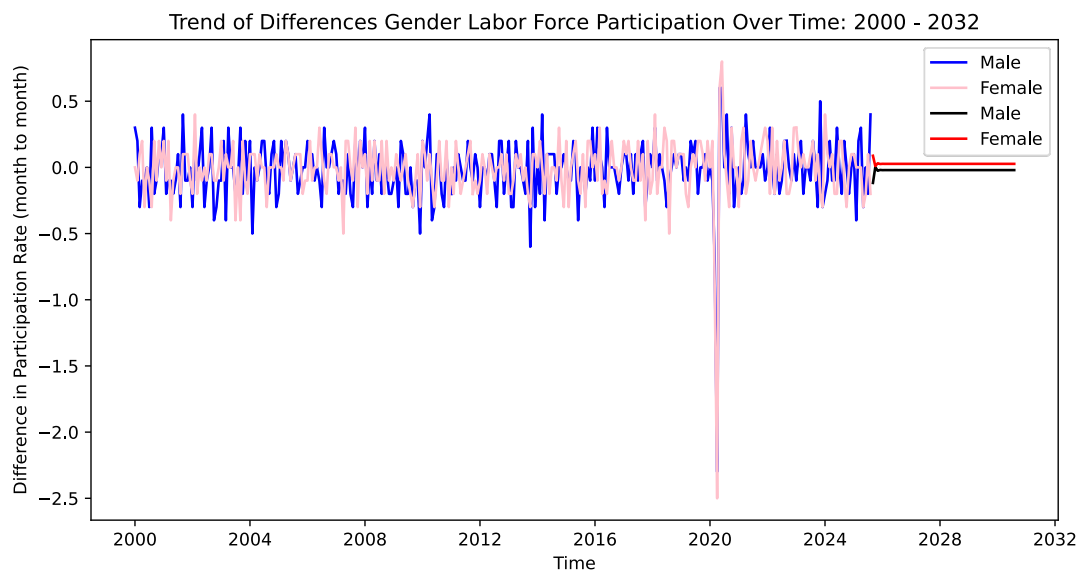
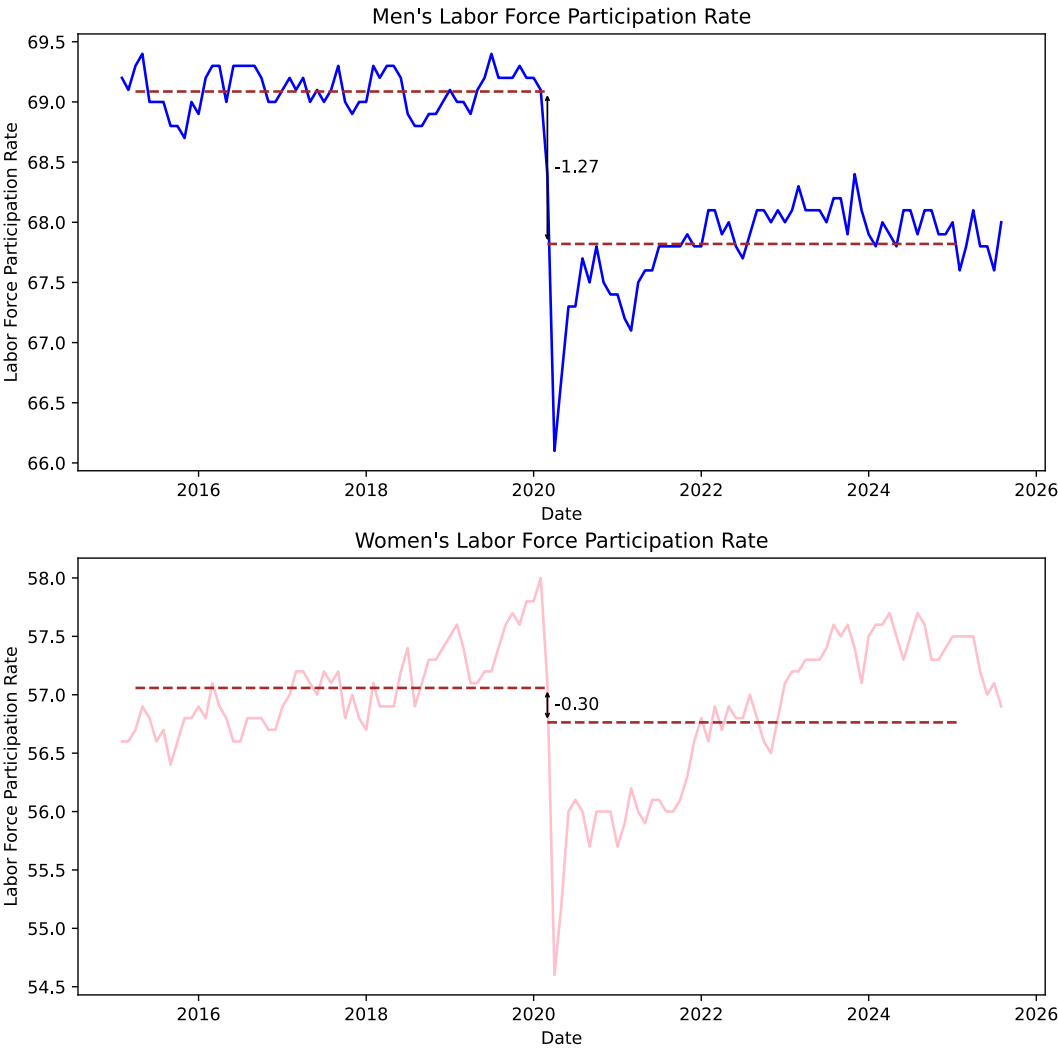


Figure 3:



Dotted lines represent the average labor force participation rate over the five years before and after COVID.

Table 1:

	Industry	coef	p-value
0	All Industries	-1.699006	4.571251e-94
1	Construction	-0.570648	8.109740e-21
2	Durable goods manufacturing	-0.180732	1.297836e-05
3	Education and health services	-1.032716	1.083943e-14
4	Financial activities	-0.498793	1.164736e-44
5	Hospitals	-7.183301	1.085244e-04
6	Information	-0.059068	1.639361e-03
7	Manufacturing	-0.233386	5.294496e-05
8	Nondurable goods manufacturing	-0.052836	3.495780e-02
9	Professional and business services	-1.523461	2.891103e-45
10	Professional and technical services	-12.439522	6.355289e-06
11	Public administration	-0.559998	4.963299e-52
12	Real estate and rental and leasing	-3.087646	4.888980e-02
13	Transportation and utilities	-0.538424	5.957658e-12

Table 2:

	Industry	coef	p-value
0	Accommodation and food services	31.625314	6.559766e-05
1	Agriculture and related industries	0.092501	2.459063e-15
2	Arts, entertainment, and recreation	14.570427	3.315552e-07
3	Educational services	10.920603	2.850981e-04
4	Health care and social assistance	19.880557	5.388884e-06
5	Health service, except hospitals	15.254989	1.345643e-07
6	Leisure and hospitality	0.723205	3.581095e-09
7	Management, administrative, and waste services	28.462852	3.923042e-17
8	Mining, quarrying, and oil and gas extraction	0.081893	4.699305e-14
9	Nonagriculture industries	138.567645	9.781431e-05
10	Other services, private households	0.077058	3.240005e-18
11	Retail trade	0.851607	3.668877e-22
12	Social assistance	11.803764	6.552757e-15
13	Utilities	1.425176	3.096439e-02
14	Wholesale and retail trade	1.008381	3.325405e-25
15	Wholesale trade	0.157063	7.229429e-13

Figure 4:

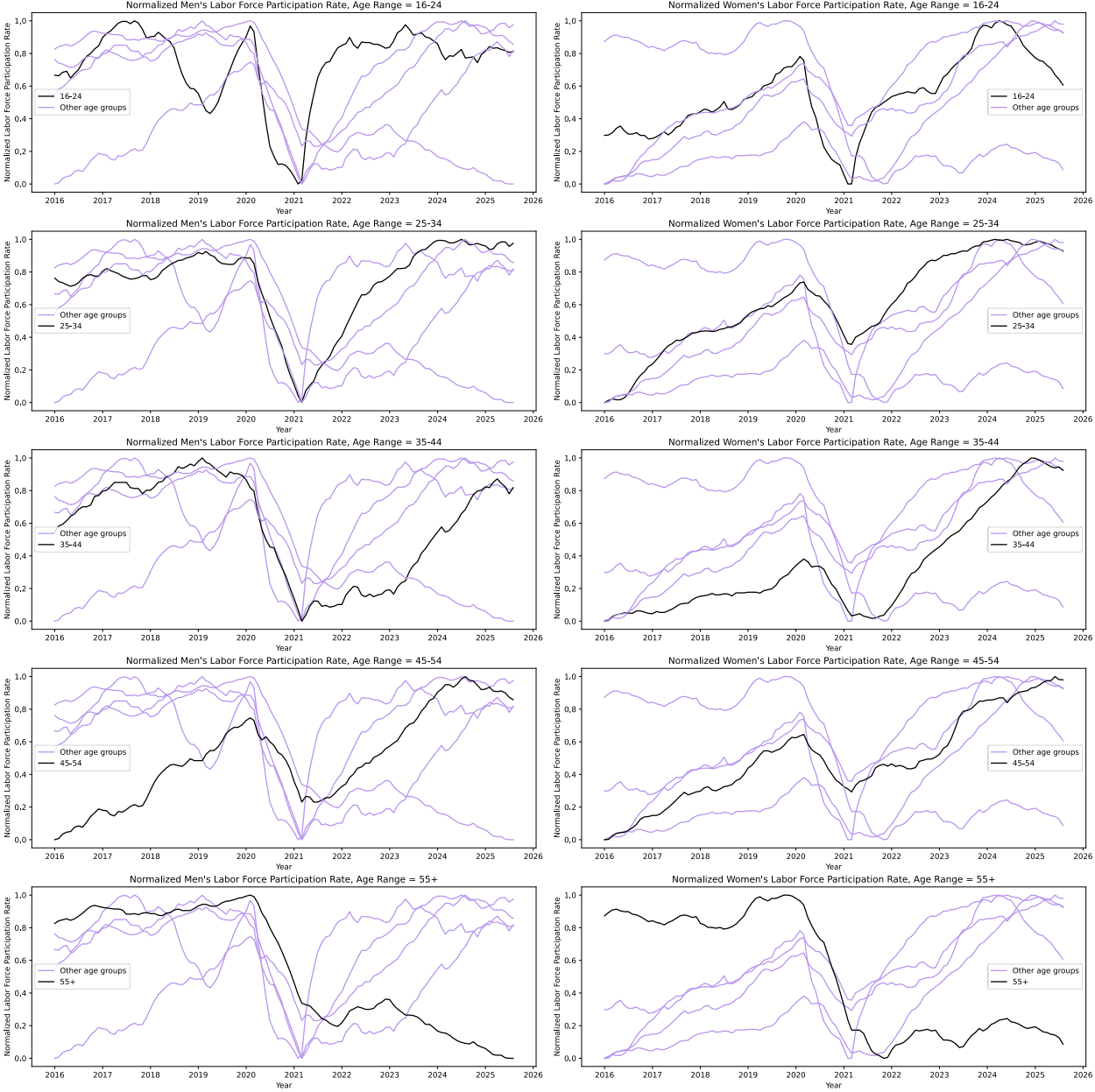
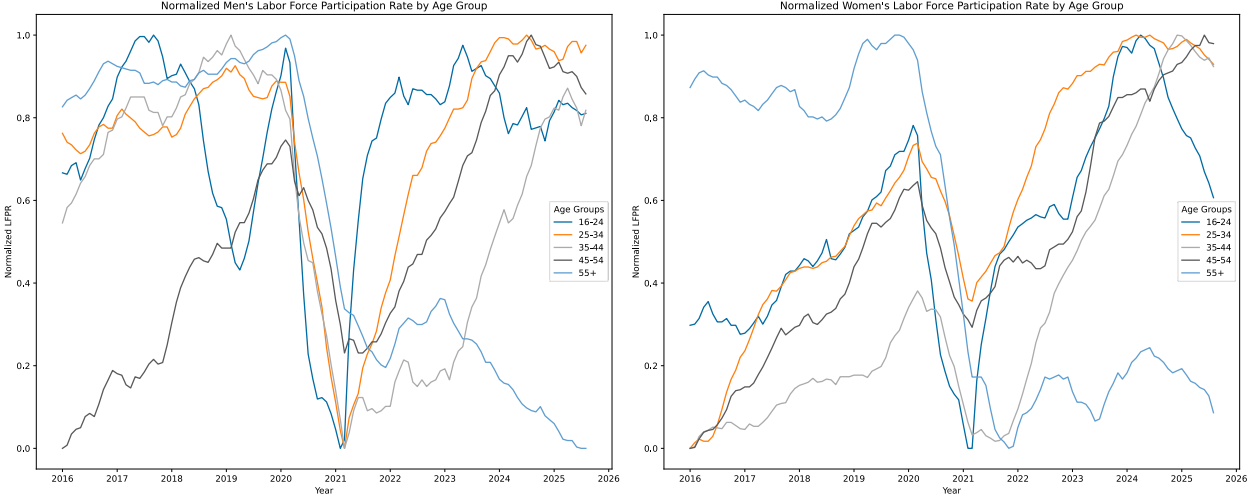


Figure 5:



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